

Corrigé — Exercice 5

$$1) \text{ a) } f'(x) = \frac{1 + \frac{x}{\sqrt{x^2+2}}}{x + \sqrt{x^2+2}} = \frac{1}{\sqrt{x^2+2}}. \quad \text{b) } I = [f(x)]_0^1 = \ln \frac{1+\sqrt{3}}{\sqrt{2}}.$$

$$2) \text{ a) } K = \int_0^1 \frac{x^2+2}{\sqrt{x^2+2}} dx = J + 2I. \quad \text{b) IPP } (u = \sqrt{x^2+2}, dv = dx) : K = [x\sqrt{x^2+2}]_0^1 - \int_0^1 \frac{x^2}{\sqrt{x^2+2}} dx = \sqrt{3} - J.$$

$$\text{c) } J + 2I = \sqrt{3} - J \Rightarrow J = \frac{\sqrt{3}}{2} - I = \frac{\sqrt{3}}{2} - \ln \frac{1+\sqrt{3}}{\sqrt{2}}, \text{ puis } K = \sqrt{3} - J = \frac{\sqrt{3}}{2} + \ln \frac{1+\sqrt{3}}{\sqrt{2}}.$$